Things to Do

- Course website
  - Homework 2 - Submission
  - Final Project

Final Project

- Final Project:
  - Register team & topic (link on course website)

  - Final Project - Grading:
    - Project Presentation: 40%
    - Project Report: 30%
    - Project Implementation (source code + demo): 30%

  - Final Project – Submission Due:
    - Project Presentation: with registered date
    - Project Report: May 7, 2026
    - Project Implementation (source code + demo): May 7, 2026

Final Project - Presentation

- Final Project Presentation (15 minutes per team):
  - Introduce the topic
  - Literature review
  - Motivation and the proposed method in details
  - <u>Your</u> ideas, extensions and improvements (if have)
  - Experimental Results & Analysis

Final Project - Report

- Final Project – Submission: THREE components

1. Presentation Slides
2. Your Source Code implementation (no middle or temp. files)
3. Report (3-4 pages at max):
   - Section 1: Introduce the topic
   - Section 2: Literature review
   - Section 3: Motivation and the proposed method in details
   - Section 4: <u>Your</u> ideas, extensions, improvements (if have)
   - Section 5: Experimental Results & Analysis
   - Section 6: Instructions how to compile/run your code

Segmentation

- Overview
- Thresholding
- Region Growing
- Split and Merge
- Watershed Segmentation
- Conclusion

Overview

- The purpose of segmentation is to divide an image into “visually sensible” regions corresponding to objects of interest
- Segmentation is the “dual” of edge detection where we focus on finding object boundaries
- In most cases, the pixels inside a region are homogeneous (same brightness or color)

Overview

- Different segmentation algorithms have been devised to group similar pixels together to form regions for each object of interest
  - Global – thresholding
  - Bottom up – region growing
  - Top down – split and merge
  - Geometric – watershed
- Once regions are found, we often need to calculate their properties (size, position, shape)

Thresholding

- The basic idea of image thresholding is to divide the intensity range in an image into two categories based on a threshold value T
  - Background = {0..T-1}
  - Object = {T..255}

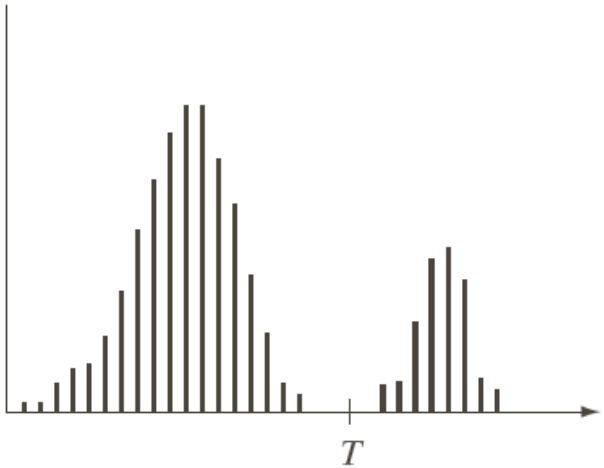
- The key is finding the threshold value T that produces the most sensible results

Thresholding

- Many interactive image processing systems allow users to select thresholds manually
- Some systems use *a priori* domain knowledge to select appropriate thresholds (eg. fax)
- Most automatic threshold selection methods are based on the intensity distribution of the input image and optimize some quality metric

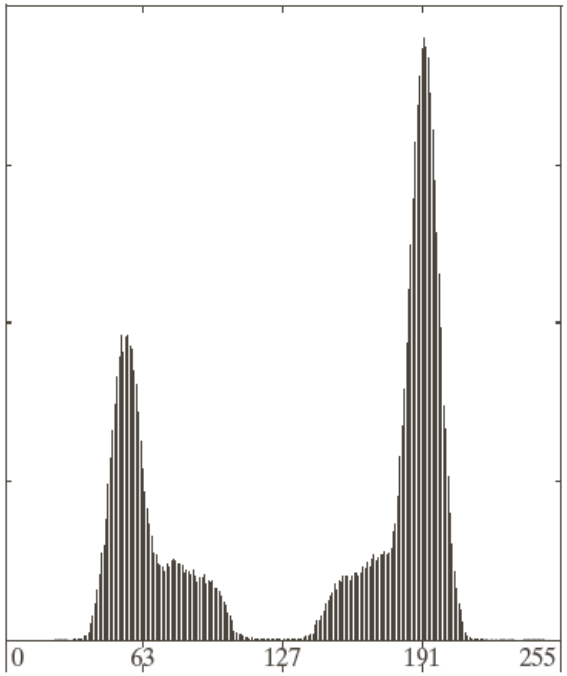
Thresholding

- The *mean midpoint threshold* algorithm selects a threshold value T to be the midpoint between the mean object intensity and the mean background intensity
  - Guess initial T
  - Calculate M_1= mean < T
  - Calculate M_2= mean > T
  - Update $T=(M_1+M_2)/2$
  - Iterate until convergence

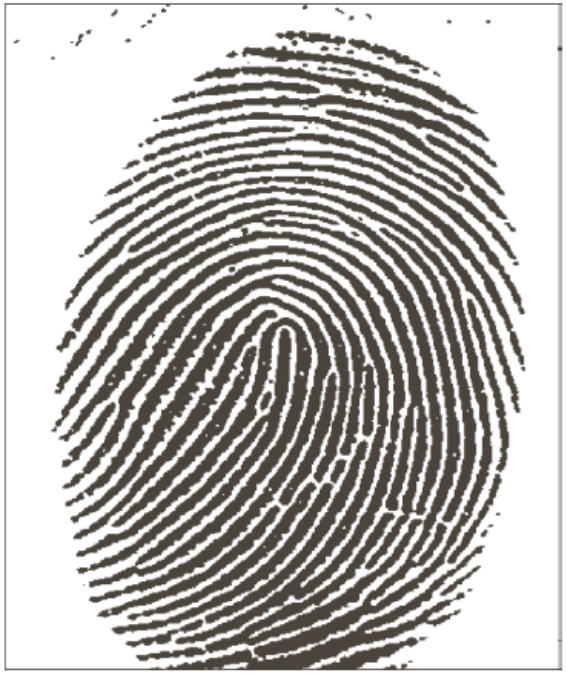


Thresholding

original image



histogram



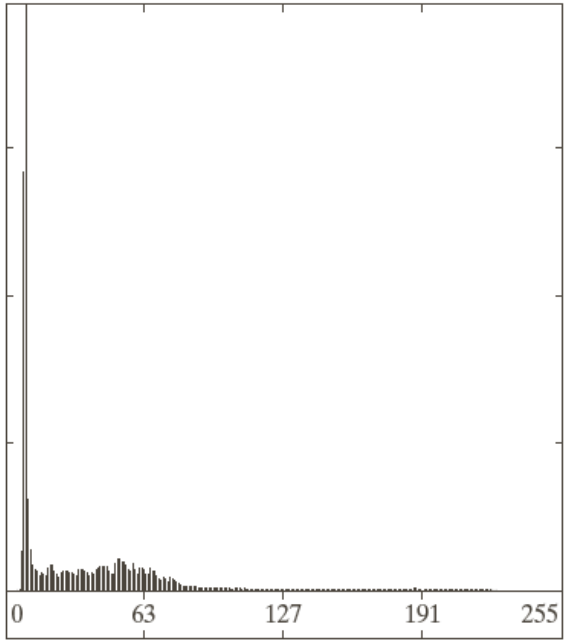
threshold image

Thresholding

- In 1979 Otsu developed a thresholding method that minimizes the *intra-class variance* and maximizes the *inter-class variance*

Thresholding

original image



histogram

threshold image

Thresholding

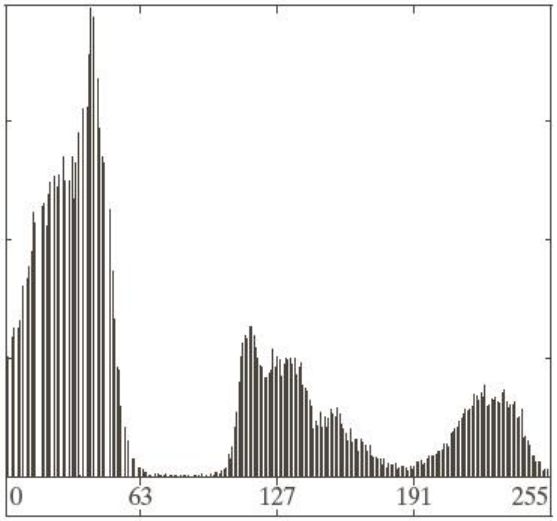
original image

threshold image

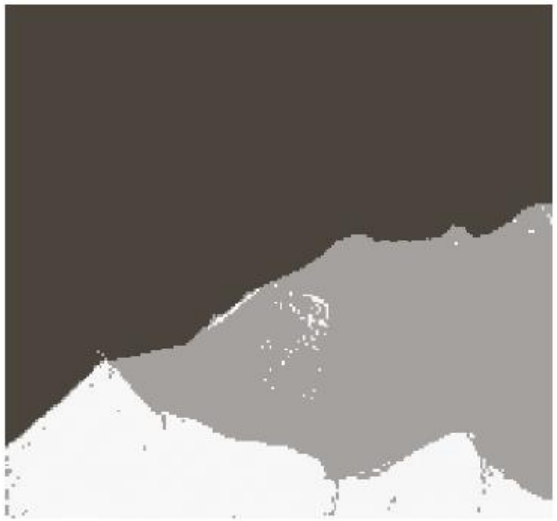
Thresholding

- Thresholding has also been extended to more than two classes using multiple thresholds

original image



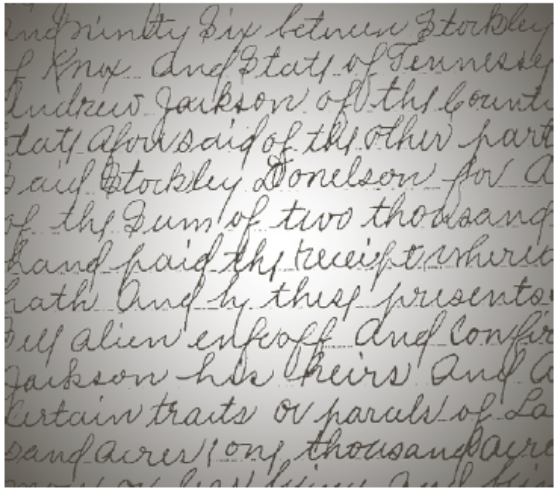
histogram



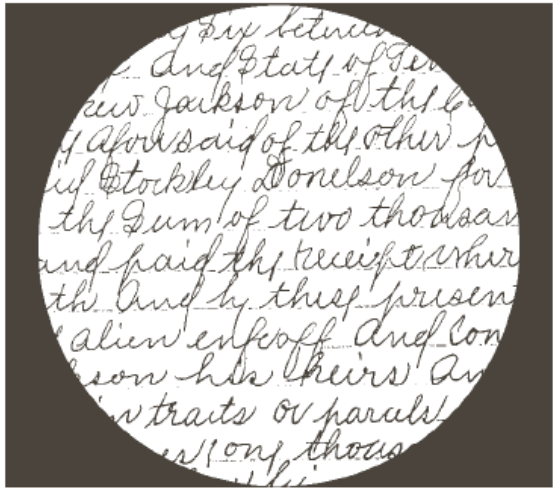
threshold image

Thresholding

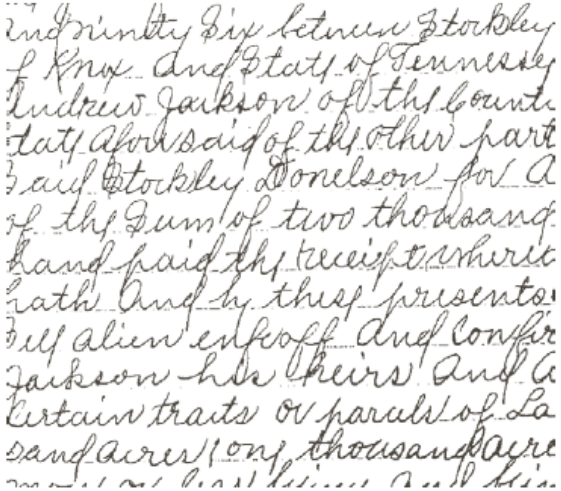
- Techniques have also been developed to deal with images with non-uniform backgrounds



original image



global threshold



local threshold

Region Growing

- Region growing is a bottom-up image segmentation approach
  - We start with a “seed point” inside an object
  - We iteratively grow a region around this seed point by adding points that are “similar” to the object
- Key questions
  - How do we select seed points?
  - How do we measure similarity?
  - When do we stop region growing process?

Region Growing

- Seed point selection options
  - Manual selection by user
  - Random selection by program
  - Automatic selection based on *a priori* knowledge of the intensity range of objects of interest

- We assume that the seed point and neighboring pixels represent the expected intensity values in the object of interest

Region Growing

- In order to grow the region we look at the N neighboring pixels and select the most similar
- Pixel similarity options

  (neighbor – seed) < threshold

  (neighbor – region mean) < threshold

  (neighbor – region mean) < region stddev
- We stop growing region when no neighbor meets the similarity requirement

Region Growing

1	0	1	0	0	1	20	19
2	1	1	1	2	22	20	18
1	0	2	7	19	3	1	1
0	1	1	2	20	22	2	0
0	0	2	21	18	19	17	2
1	3	16	19	22	20	2	1
1	16	17	16	2	2	1	1
0	0	18	17	1	0	0	0

original image with seed point

Region Growing

1	0	1	0	0	1	20	19
2	1	1	1	2	22	20	18
1	0	2	7	19	3	1	1
0	1	1	2	20	22	2	0
0	0	2	21	18	19	17	2
1	3	16	19	22	20	2	1
1	16	17	16	2	2	1	1
0	0	18	17	1	0	0	0

4 neighbors of seed point

Region Growing

1	0	1	0	0	1	20	19
2	1	1	1	2	22	20	18
1	0	2	7	19	3	1	1
0	1	1	2	20	22	2	0
0	0	2	21	18	19	17	2
1	3	16	19	22	20	2	1
1	16	17	16	2	2	1	1
0	0	18	17	1	0	0	0

updated region and neighbors

Region Growing

1	0	1	0	0	1	20	19
2	1	1	1	2	22	20	18
1	0	2	7	19	3	1	1
0	1	1	2	20	22	2	0
0	0	2	21	18	19	17	2
1	3	16	19	22	20	2	1
1	16	17	16	2	2	1	1
0	0	18	17	1	0	0	0

updated region and neighbors

Region Growing

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updated region and neighbors

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updated region and neighbors

Region Growing

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updated region and neighbors

Region Growing

1	0	1	0	0	1	20	19
2	1	1	1	2	22	20	18
1	0	2	7	19	3	1	1
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0	0	2	21	18	19	17	2
1	3	16	19	22	20	2	1
1	16	17	16	2	2	1	1
0	0	18	17	1	0	0	0

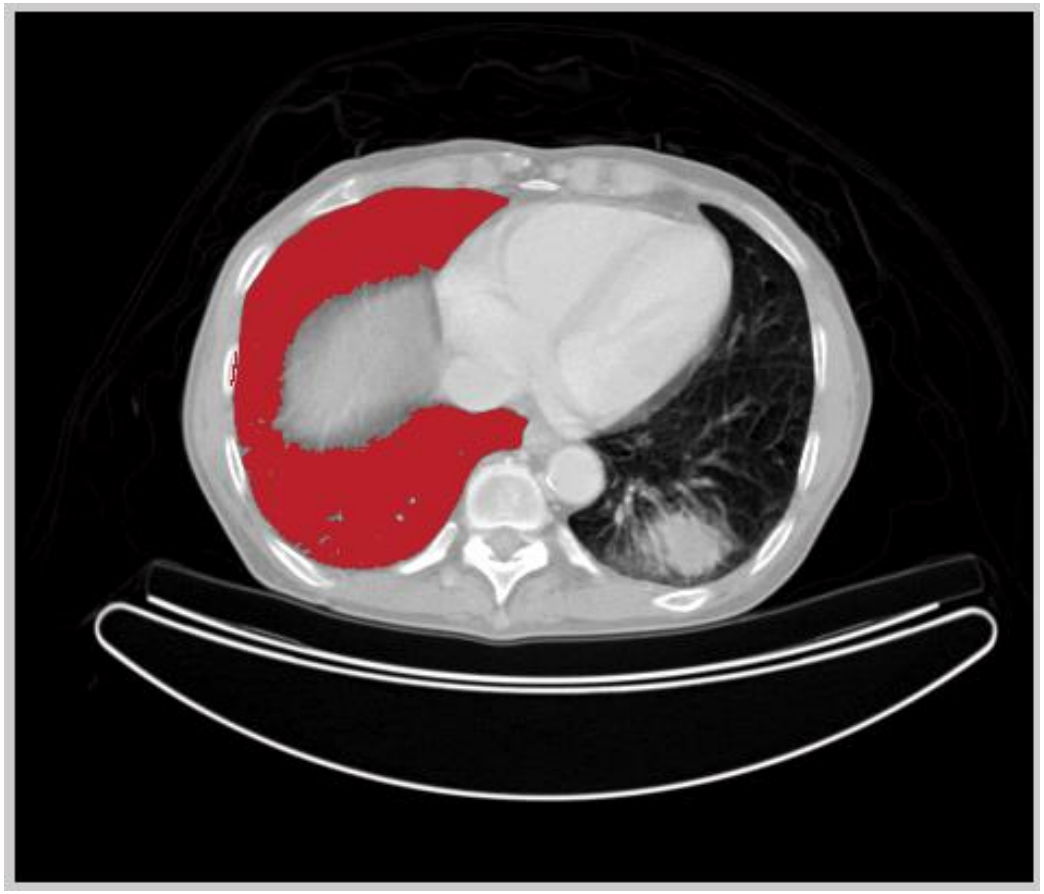
final region and neighbors

Region Growing

- Region growing works well for images with homogeneous object intensities and uniform background intensities

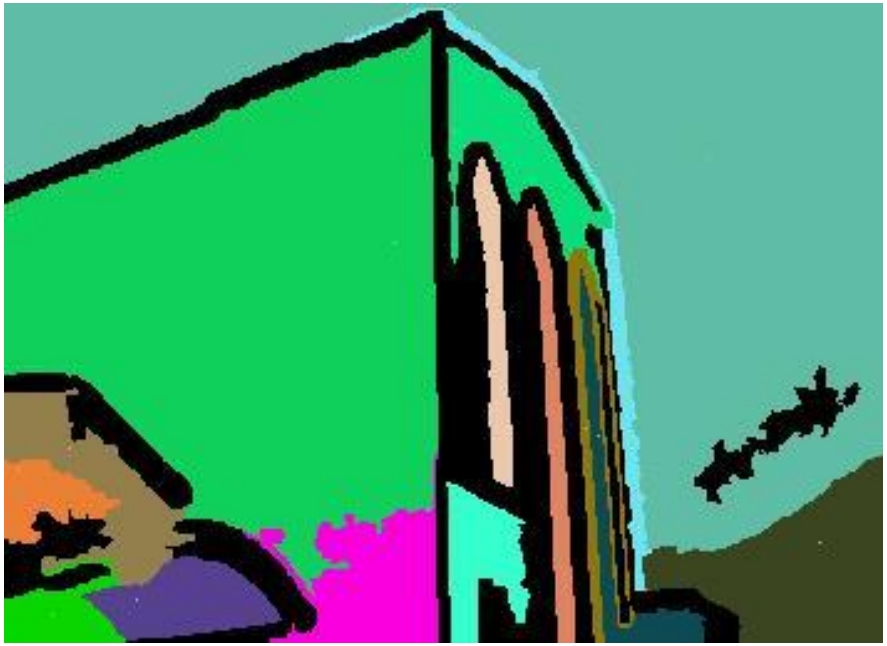
- Issues with region growing
  - Starting with different seed points within an object may result in slightly different regions
  - Using 4-neighbors instead of 8-neighbors may result in disconnected object regions

Region Growing



manual seed selection

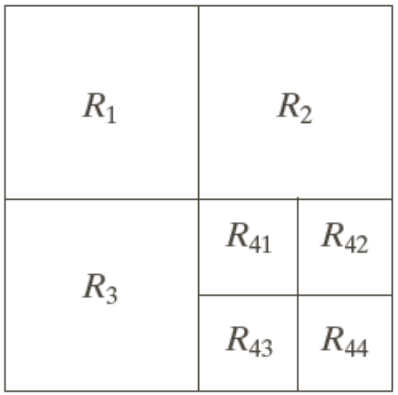
Region Growing



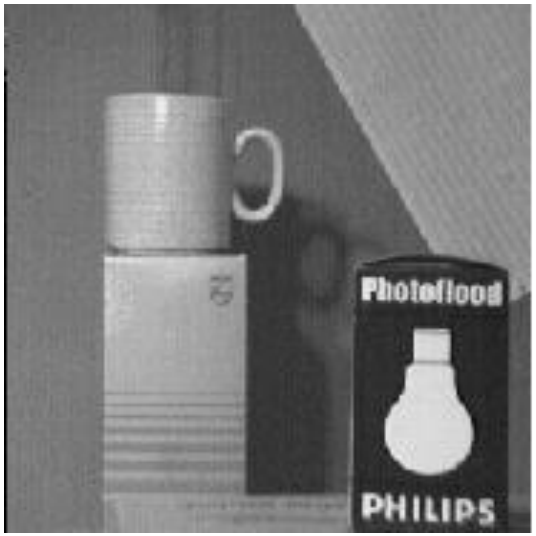
multiple random seed selection

Split and Merge

- Split and merge is a top-down image segmentation method
  - Divide the image into 4 regions
  - Recursively subdivide regions if they are not homogeneous (eg when intensity range or variance is large)
  - When finished, merge all adjacent regions that are similar to each other (eg mean values differ by less than the region variance)



Split and Merge



original image

after split

after merge

Split and Merge

original image

after split and merge

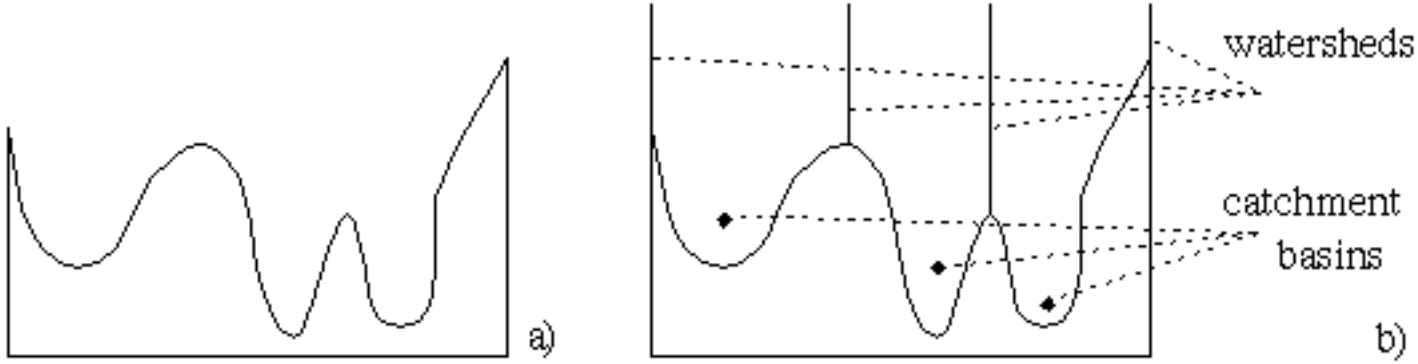
Split and Merge

original image

after split and merge

Watershed Segmentation

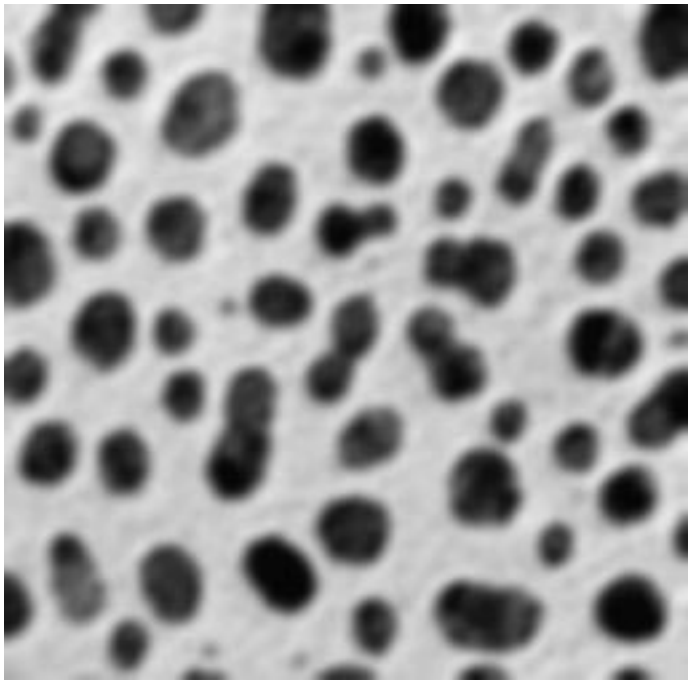
- The idea behind watershed segmentation is to use the image intensity or image gradient to define topological regions corresponding to *catchment basins*



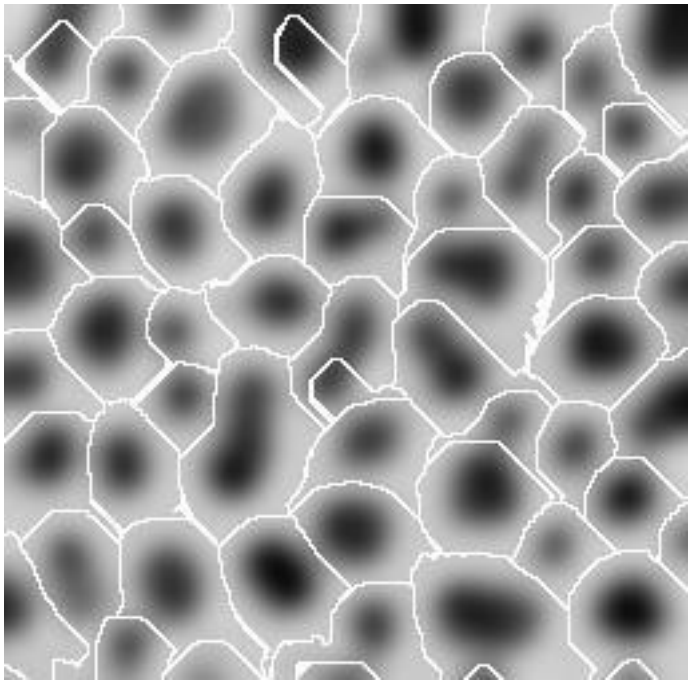
Watershed Segmentation

- A variety of methods have been developed to calculate watersheds
  - Some methods “flood” the image from local minima and keep track of when water from one catchment basin merges with another
  - Other methods “simulate rainfall” by walking downhill from each pixel and classify points based on which local minima eventually reach

Watershed Segmentation



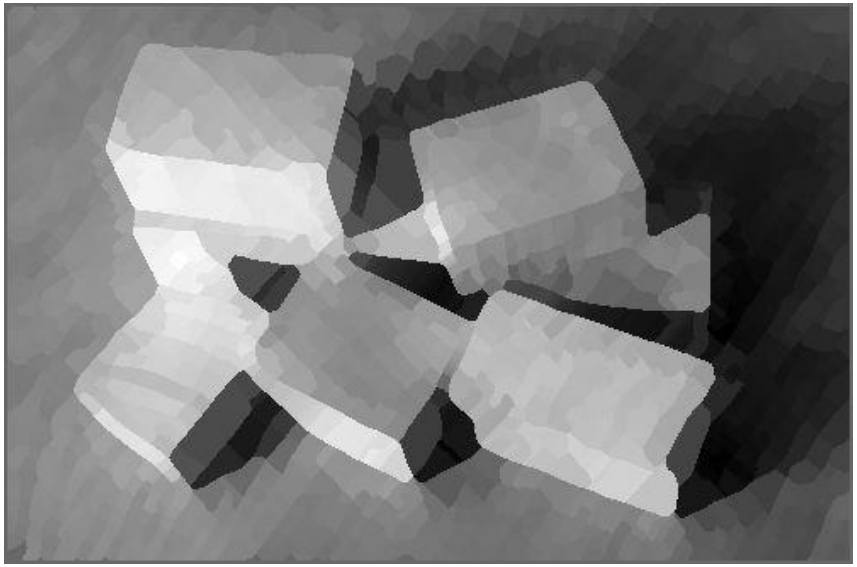
original image



watershed boundaries

Watershed Segmentation

original image



watershed regions

Conclusion

- Image segmentation is one of the oldest and most studied areas of computer vision
- A wide variety of techniques have been invented for image segmentation
  - Different types of input images
  - Different output types (edges vs regions)
  - Different speed / accuracy requirements